

YELP : Reviews & Sentiment Analysis

Business Queries :

1. Find number of businesses in each category

```
with cte as(
select
    business_id,
    trim(A.value) as category
from tbl_yelp_biz,
LATERAL split_to_table(categories,',') A
)
select
    category,
    count(category) as n_business
from cte
group by category
order by n_business desc;
```

	A CATEGORY	# N_BUSINESS
1	Restaurants	52268
2	Food	27781
3	Shopping	24395
4	Home Services	14356
5	Beauty & Spas	14292
6	Nightlife	12281
7	Health & Medical	11890
8	Local Services	11198
9	Bars	11065
10	Automotive	10773

2. Top 10 users who have reviewed "Restaurants"

```
select
  r.user_id,
  count(DISTINCT r.business_id)
from tbl_review r
join tbl_yelp_biz b
on r.business_id = b.business_id
where b.categories ilike '%restaurants'
group by 1
order by 2 desc
limit 10;
```

	A USER_ID	# COUNT(DISTINCT R.BUSINESS_ID)
1	-G7Zk1wlWBBmD0KRy_sCw	290
2	_BcWyKQL16ndpBdggh2kNA	284
3	1HM81n6n4IPIFU5d2Lokhw	233
4	fr1Hz2acAb3OaL3l6DyKNg	231
5	ET8n-r7glWYqZhuR6GcdNw	208
6	Xw7ZjaGfr0WNVt6s_5KZfA	196
7	pou3BbKslozfH50rxmnMew	186
8	bYENop4BuQepBjM1-BI3fA	182
9	vmUqcqMjIWoBM6qfmUXgyQ	182
10	ppsm8EDKJA1fp1yTCP3RrQ	176

3. Find the most popular categories of business based on number of reviews

```
with cte as(
select
  business_id,
  trim(A.value) as category
from tbl_yelp_biz,
LATERAL split_to_table(categories,',') A
)
select
  category,
  count(*) as n_reviews
from cte
join tbl_review r
on cte.business_id = r.business_id
group by 1
order by 2 desc;
```

	🔍 CATEGORY	# N_REVIEWS
1	Restaurants	4724471
2	Food	1813593
3	Nightlife	1539757
4	Bars	1455553
5	American (Traditional)	1011646
6	American (New)	984540
7	Breakfast & Brunch	867430
8	Sandwiches	691864
9	Seafood	620247
10	Event Planning & Services	609553

4. Find the 3 most recent reviews for each business

with biz_rank as(

select

*,

row_number() over (partition by business_id order by review_date desc) rn

from tbl_review

)

select business_id,review_date

from biz_rank

where rn <=3;

	🔍 BUSINESS_ID	⋮ ↑	📅 REVIEW_DATE
1	OP3GwDnorIKLItqZYdm4dA	Workspaces	2020-08-25
2	OP3GwDnorIKLItqZYdm4dA		2020-07-29
3	OP3GwDnorIKLItqZYdm4dA		2019-02-03
4	WvgvsSWK2yNf_ZigaDZmlw		2019-08-02
5	WvgvsSWK2yNf_ZigaDZmlw		2018-11-30
6	WvgvsSWK2yNf_ZigaDZmlw		2018-01-19
7	JrIYnMIFLq2NWLk9EbM4AA		2020-10-11
8	JrIYnMIFLq2NWLk9EbM4AA		2020-07-12
9	JrIYnMIFLq2NWLk9EbM4AA		2019-09-30

5. Find the month with highest number of reviews

```
select
    month(review_date) as review_month,
    count(*) as n_reviews
from tbl_review
group by 1
order by 2 desc;
```

id	# REVIEW_MONTH	# N_REVIEWS
1	7	654627
2	8	636384
3	1	604532
4	6	601737
5	3	598555
6	5	586575
7	10	571809
8	9	565374
9	4	551471
10	2	544125
11	12	543573
12	11	531518

6. Find percent of 5-star rating for each business

```
with rating as(
    select
        business_id,
        sum(case when stars = 5 then 1 else 0 end) as star5_ratings,
        count(*) as total_ratings
    from tbl_review
    group by 1
)
select
    business_id,
    star5_ratings,
    total_ratings,
    (100.0 * star5_ratings / total_ratings) as star5_pct
from rating;
```

id	BUSINESS_ID	# STAR5_RATINGS	# TOTAL_RATINGS	# STAR5_PCT
1	tGQytMbOXDdZS_OSK9hkBw	90	151	59.602649
2	EF_9W5s9IFSjv4forNxwng	202	328	61.585366
3	yEVRdb-9sBqzRiPC5BxwWw	275	425	64.705882
4	BOjpe0u5u_cm9SWMS2KLFg	485	742	65.363881
5	j-qtD55OLfSqfsWuQTDJg	1320	2884	45.769764
6	yvBDnfgVQy_hCDGj3Hy00w	72	318	22.641509
7	dsfRnIRgfbDJC8os848B6A	1959	3028	64.696169
8	SA10MB4CEw0YikMuhOxyYw	251	571	43.957968
9	i2WiqWwv3H9DZJ8uaqwPDA	83	296	28.040541
10	aunmz06iWvo3bd6MMHEbqg	18	313	5.750799

7. Find top 5 most reviewed business in each city

with cte as (

select

business_id,

review_count,

city

from tbl_yelp_biz

)

select *

from cte

qualify dense_rank() over (partition by city order by review_count desc) <= 5;

#	BUSINESS_ID	REVIEW_COUNT	CITY
1	EAc4F8EA78CwJtmE_hC4mw	5	elmwood
2	PvHhnbfr1HBEdIJuBqNvag	98	St. Pete Beach
3	k-tHn9uxgWdq1ROHqwc-lw	965	St. Pete Beach
4	b1Rz5EI0mSHkOfab5J8P0g	94	St. Pete Beach
5	aoRv_tZfoGHjXkPqy94ryQ	93	St. Pete Beach
6	RLVuL5T2uNCP6bHvDxMkOQ	92	St. Pete Beach
7	c4cuo-lvHZBbREVUgndBcQ	5	Valencia West
8	AsPLgxF7AZm7VTmZ9Ui6YA	19	Maryland Height
9	wUZxm4KN2wqlfkvXVZXUw	998	Nashville
10	zJmp_J8SUNosLxk6borpFg	993	Nashville
11	IJEivHHf5A8LzXXOn0o0Nw	993	Nashville
12	3CFVBCfjdCvESS1ogBv21A	992	Nashville
13	KFJUPBFX9DCCfkIEPjQJAg	99	Nashville

8. Find avg. rating of businesses that have at least 100 reviews

select

r.business_id,

count(*) as total_reviews,

round(avg(r.stars),2) as avr_rating

from tbl_review r

join tbl_yelp_biz b

on r.business_id = b.business_id

group by 1

having total_reviews >=100;

#	BUSINESS_ID	TOTAL_REVIEWS	AVR_RATING
1	VAeEXLbEcl9Emt9KGYq9aA	3105	4.04
2	gebiRewfieSdtt17PTW6Zg	502	3.39
3	AGlh4ZDv6jnoiYfz7At9mw	2778	3.88
4	RQAF6a0akMiot5IZznMNNw	2725	4.21
5	wz8ZPfySQczcPgSyd33-HQ	3634	2.98

9. List of top 10 reviewers ,along with businesses they reviewed

```
with user_cte as(
  select
    user_id,
    count(*) as total_reviews
  from tbl_review
  group by 1
  order by 2 desc
  limit 10
)
select
  distinct user_id,
  business_id
from tbl_review
where user_id in (select user_id from user_cte)
order by 1;
```

10. Find top 10 businesses with highest 'Positive' sentiment reviews

```
select
  business_id,
  count(*) as total_positive_reviews
from tbl_review
where sentiments = 'Positive'
group by 1
order by 2 desc
limit 10;
```

id	BUSINESS_ID	TOTAL_POSITIVE_REVIEWS
1	_ab50qdWok0DdB6XOrBitw	7558
2	ac1AeYqs8Z4_e2X5M3if2A	7349
3	GXFMD0Z4jEVZBCsbPf4CTQ	5972
4	ytynqOUb3hjKeJfRj5Tshw	5739
5	oBNrLz4EDhiscSibOI8uAw	5148
6	iSRTaT9WngzB8JJ2YKJUig	4916
7	_C7QIQQc47AOEv4PE3Kong	4911
8	VQcCL9PINL_wkGf-uF3fjg	4883
9	GBTPC53ZrG1ZBY3DT8Mbcw	4587
10	6a4gLLFSgr-Q6CZXLzBGQ	4410